

SEMICONDUCTOR IN NANODEVICES

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INTRODUCTION

As we noticed, semiconductors have become **indispensable** in modern life, especially in nanodevices, and are now integrated into a vast range of devices we use **every day**—from smartphones and laptops to advanced medical systems and smart infrastructure. Semiconductors form the **technological backbone** of modern innovation, and our role as nanotechnology engineers is to provide the knowledge and understanding needed to **explore and advance** this critical field.



NANOSCALE INNOVATION

At the nanoscale, materials exhibit **extraordinary properties** that differ from their bulk counterparts. **Quantum confinement**, **tunable bandgaps**, and **enhanced charge carrier mobility** enable semiconductors to power next-generation technologies.

UNIQUE PROPERTIES ENABLE INNOVATIVE APPLICATIONS



QUANTUM CONFINEMENT

Enables tunable electronic and optical behavior.



TUNABLE BANDGAPS

Allow light absorption and emission control.



ENHANCED CHARGE CARRIER MOBILITY

Improves speed and energy efficiency.



HIGH CHEMICAL STABILITY

Ensures reliability in harsh and diverse conditions.



THERMAL MANAGEMENT

Supports better heat dissipation and device performance.

OUR GOAL



To precisely **design, control**, and **integrate** semiconductor nanostructures, unlocking breakthroughs in **electronics, photonics, quantum technologies**, and beyond.



This understanding empowers us to drive innovation in nanomedicine, flexible electronics, energy conversion, and neuromorphic computing—shaping a **smarter, faster, and more sustainable future**.

WHAT WE ARE GOING TO DO IN OUR PRESENTATION



fundamental physics of semiconductor

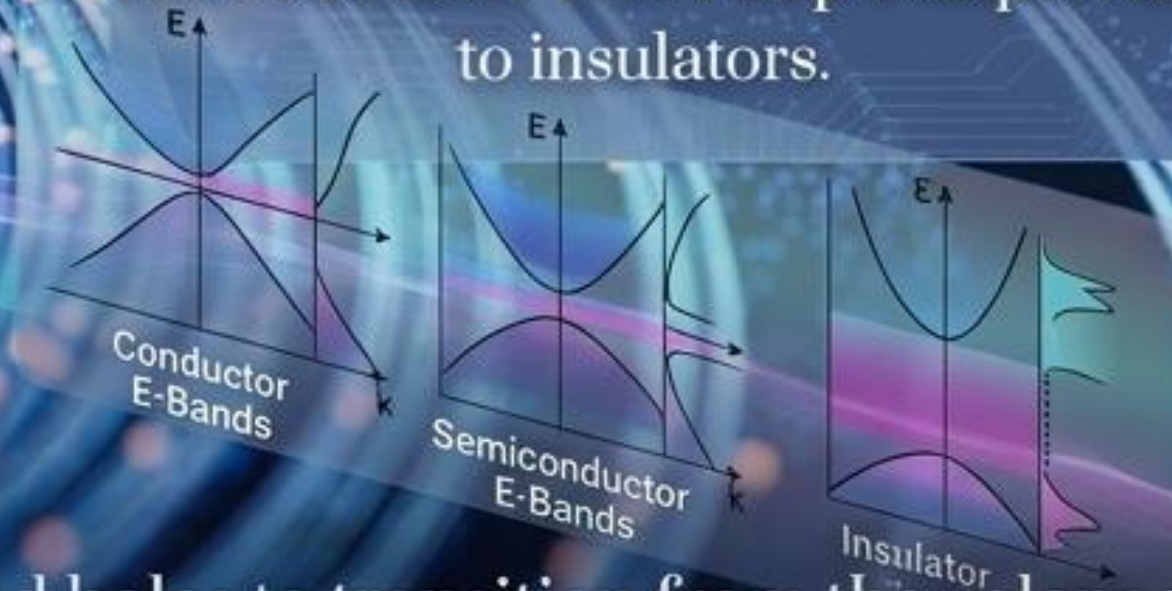
First, we need to analyze the difference in the structure of different materials.

$$E = h\nu = \frac{1}{1/\pi} \frac{1}{2\pi\epsilon_0}$$

Conductors: Materials that have a loosely bound valence electrons (No Band Gap).

-Semiconductors: Materials that have a smaller Band Gap compared to insulators.

Insulators: Materials that have a tightly bound valence electrons (Huge Band Gap).

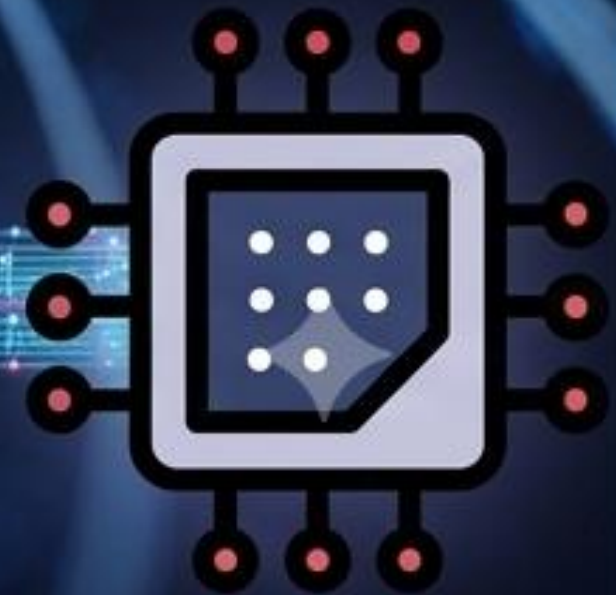


What are Energy Bands?

$$E = h\nu = \frac{2m\nu}{\hbar} + (l)\hbar\omega$$
$$= \exp\left(\frac{\mathcal{P}^2}{2m} + \frac{1}{\hbar\omega}\right)$$

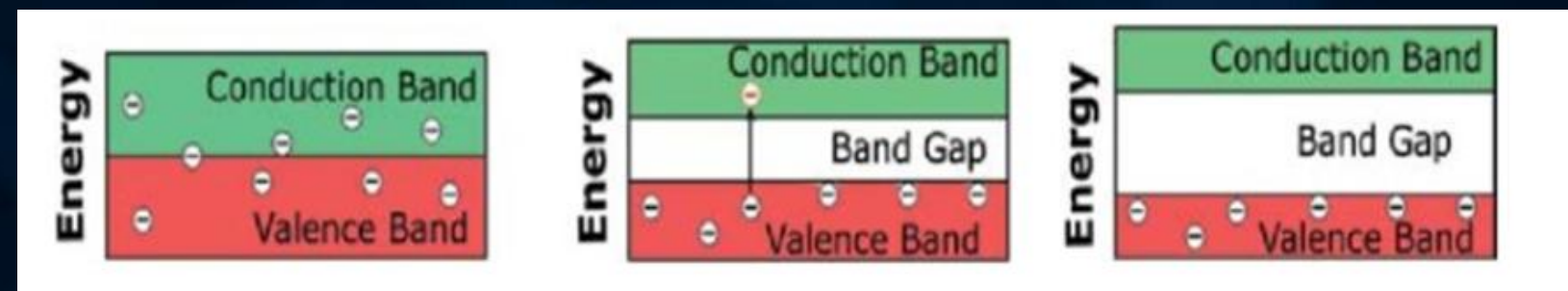
E = Fermi-Dirac function

The energy required for electrons and holes to transition from the valence band to the conduction band is called a band gap. A semiconductor with a large value is called a wide-band-gap semiconductor. They have a moderate band gap, which allows controlled conductivity — making them perfect for electronic devices.



WHAT EFFECT DO BAND GAPS HAVE ON CONDUCTIVITY?

From the figure below, we can see that for conduction to happen, electrons need to move from valence bands to conduction bands, therefore, band gaps determine material conduction properties.



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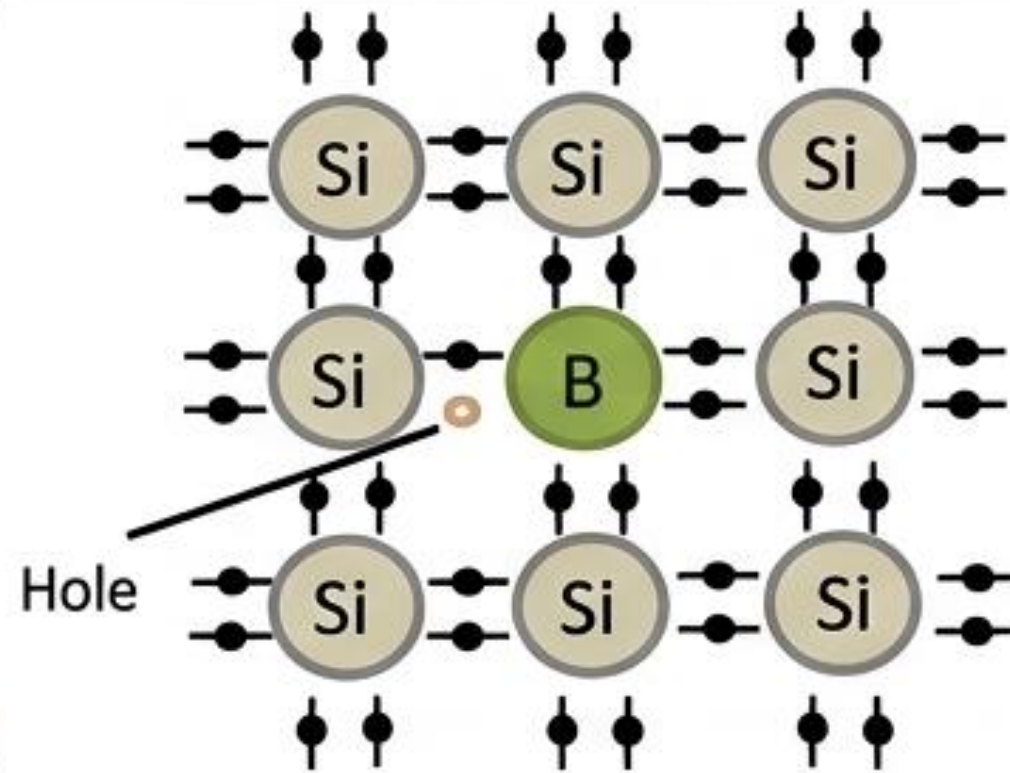
How?

Band Gaps-->>.Energy -->>Less Conductivity
(The more energy is required, the less conductivity the material is)



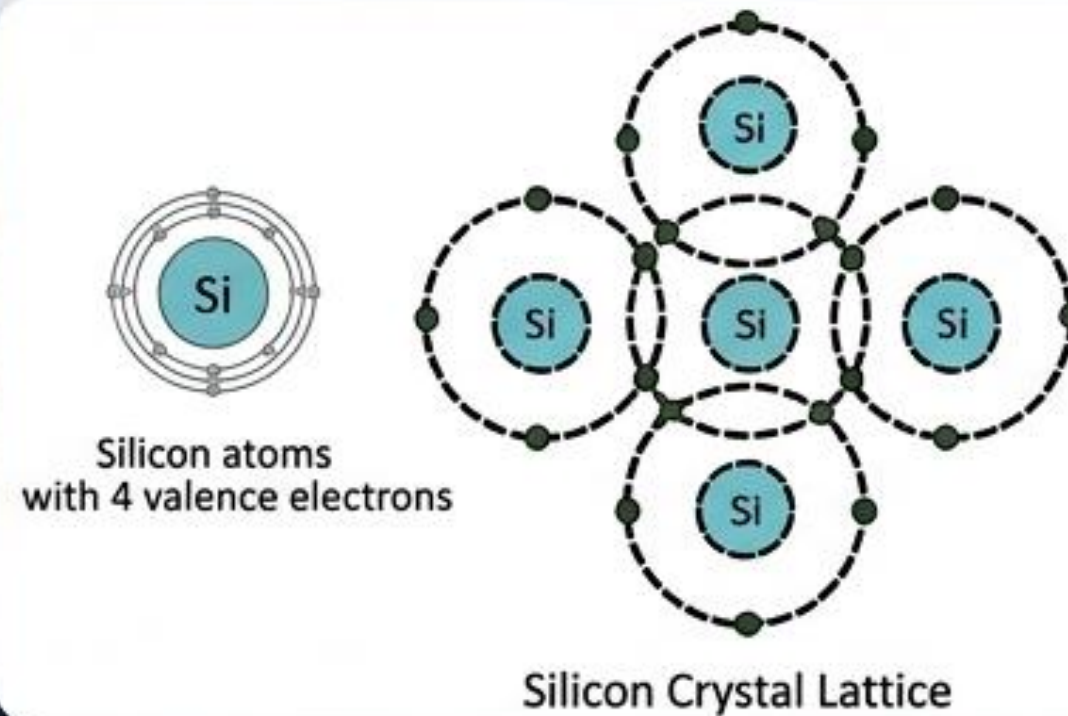
SEMICONDUCTOR FUNDAMENTALS: CRYSTAL STRUCTURE, CARRIERS, & DOPING

WHAT ARE CHARGE CARRIERS?



- Charge Carriers are particles that carry electric charge and move freely within the semiconductor material.
- *Two types:*
 - **Electrons:** Negatively charged particles that orbit the nucleus.
 - **Holes:** Represent the absence of electrons, carrying a positive charge equal in magnitude to the electronic charge.

INTRINSIC SEMICONDUCTORS & LATTICE

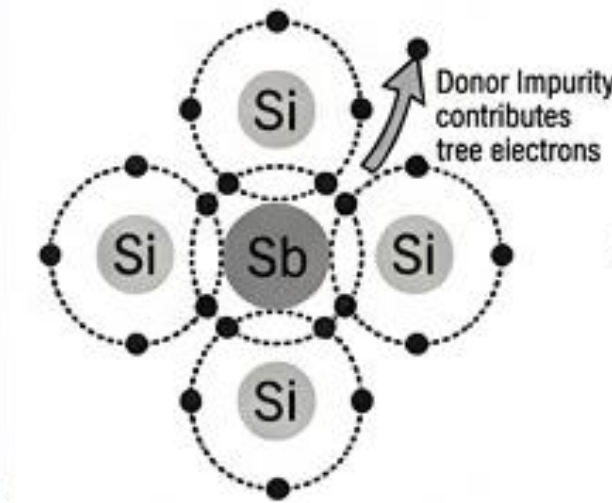


Intrinsic Semiconductors are pure materials, in other words, free from impurities.

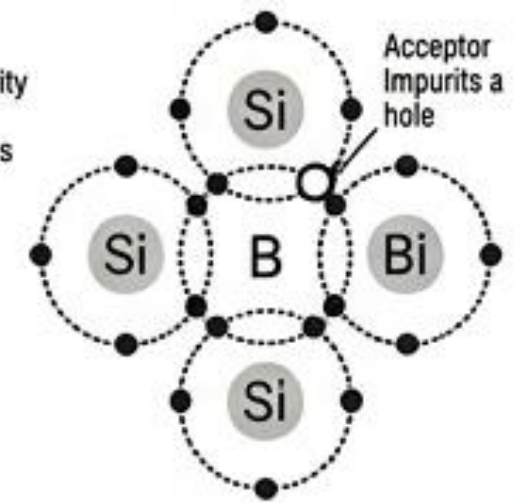
In contrast, **Extrinsic Semiconductors** are semiconductors that have been doped with impurities to change their electrical properties.

DOPING PROCESS & TYPES

N-type: (Si + As) or (Si + P)



P-type: (Si + B)



- **What is doping?** Doping is the process of adding specific impurities (or more) to a semiconductor to change its electrical properties and increase conductivity.
- Key semiconductor types depend on the impurity:
 - **Trivalent atoms** (e.g., Boron): The resultant semiconductor is **P-Type** (more positive charge carriers, *holes*).
 - **Pentavalent atoms** (e.g., Phosphorus): The resultant semiconductor is **N-Type** (more negative charge carriers, *electrons*).
- (Note on each type: each has a clear description from <IMAGE 0! >).

MATERIALS:



14
Silicon

1. SILICON (Si):

- **Use:** Most widely used material in electronics (e.g., microchips, solar cells).
- **Limitation:** Slower electron mobility compared to some newer materials.



31
Gallium
Arsenide

2. GALLIUM ARSENIDE (GaAs):

- **Use:** High-speed electronics, LEDs, and microwave frequency devices.
- **Limitation:** More expensive and brittle than silicon.



6 Graphene

3. GRAPHENE:

- Use: Advanced electronics, sensors, flexible displays, and batteries.
- Limitation: No natural band gap (difficult to “turn off” current).



42
Molybdenum
Disulfide

4. MOLYBDENUM DISULFIDE (MoS₂):

- Use: Next-gen transistors, photodetectors, flexible electronics.
- Limitation: Still in early development; less mature than silicon.

WHY CHOOSING MATERIAL IS CRUCIAL

At the nanoscale, the performance of a device heavily depends on the material it is made from. Each application (transistor, sensor, solar cell, etc.) has unique requirements, so engineers must select materials based on specific properties such as:



Electron Mobility



Band Gap



Flexibility



Chemical Stability



Thermal Conductivity

PERIODIC TABLE (RELEVANT SEMICONDUCTOR MATERIALS)

PERIODIC TABLE (RELEVANT SEMICONDUCTOR MATERIALS)										VIII A
		IIIA	IVA	VA	VIA	VIIA	4			
		5	6	7	8	9	10			
		B	C	N	O	F	Ne			
		10.811	12.011	14.007	15.999	18.998	20.183			
		13	14	15	16	17	18			
		Al	Si	P	S	Cl	Ar			
		26.982	28.086	30.974	32.06	35.453	39.948			
		29	30	31	32	33	34	35	36	
		Cu	Zn	Ga	Ge	As	Se	Br	Kr	
		63.54	65.37	69.72	72.59	74.922	78.96	79.909	83.80	
		47	48	49	50	51	52	53	54	
		Ag	Cd	In	Sn	Sb	Te	I	Xe	
		107.87	112.40	114.82	118.69	121.75	127.60	126.90	131.30	
		79	80	81	82	83	84	85	86	
		Au	Hg	Tl	Pb	Bi	Po	At	Rn	
		196.97	200.59	204.37	207.19	208.98	(210)	(210)	(222)	



KEY TAKEAWAY

The right semiconductor material enables faster, smarter, more efficient, and more reliable nano-devices that power today and shape tomorrow.



MATERIALS DRIVE INNOVATION. THE FUTURE IS BUILT AT THE NANOSCALE.

EXAMPLES OF SOME USES AND MATERIAL SELECTION::



Transistors (used in processors, memory, logic circuits):

Material Choices:

- Silicon (Si): Industry standard due to good performance and low cost.
- MoS₂ (Molybdenum disulfide): 2D material with a natural band gap and excellent scaling potential.



Photodetectors / Solar Cells:

- Material Choices:
 - Gallium Arsenide (GaAs): High efficiency for solar conversion.
 - Quantum dots (e.g., CdSe): Adjustable band gap based on particle size.



3-Flexible Electronics (used in wearable tech, foldable screens):

Material Choices:

Graphene: Extremely strong and flexible with excellent conductivity.
Organic semiconductors: Low-cost and printable on flexible surfaces.



4-Sensors (gas sensors, biosensors):

- Material Choices:
 - Graphene and MoS₂: Large surface area, sensitive to molecule interactions
 - Carbon Nanotubes: Excellent for detecting gases due to high surface-to-volume ratio.

• COMPARE MATERIAL PROPERTIES RELEVANT TO NANOSCALE PERFORMANCE:

Mobility :: is a measure of the ability of an electron or a hole to get around inside a semiconductor. It is a “constant” for a particular situation a “situation” meaning a particular temperature and doping concentration

Examples:

Silicon:

Band Gap ≈ 1.1 eV $\rightarrow$ Ideal for making transistors and circuits.

Germanium:

Band Gap ≈ 0.66 eV $\rightarrow$ Conducts more easily than silicon.

Flexibility:

What is it?

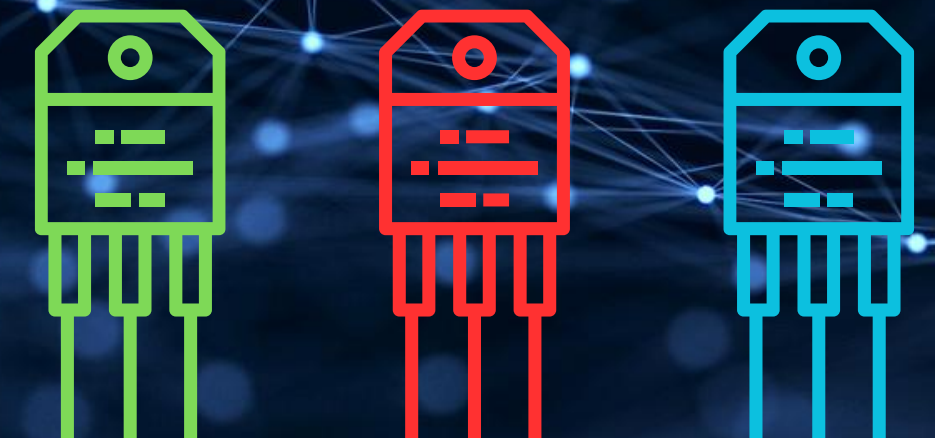
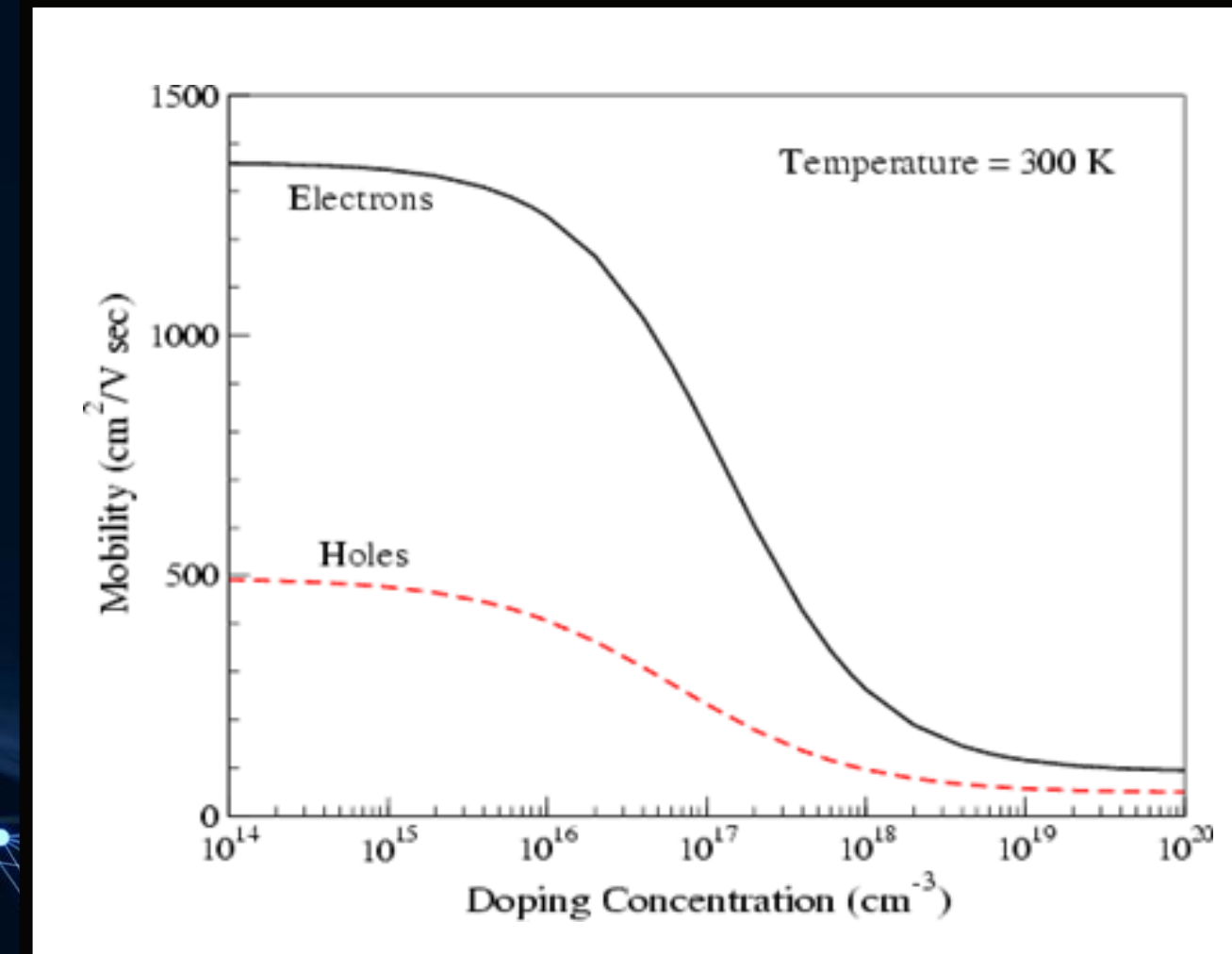
The ability of a material to bend or deform without breaking.

Why is it important?

It's essential in flexible electronics, wearables, and foldable displays.

At the nanoscale:

Materials such as carbon nanotubes and graphene combine flexibility with high strength, which is a rare and valuable combination.



FABRICATION METHODS FOR SEMICONDUCTOR NANODEVICES

1. Advanced Lithography Techniques

Lithography is one of the most important industrial methods for nano fabrication in the modern era.

Photolithography

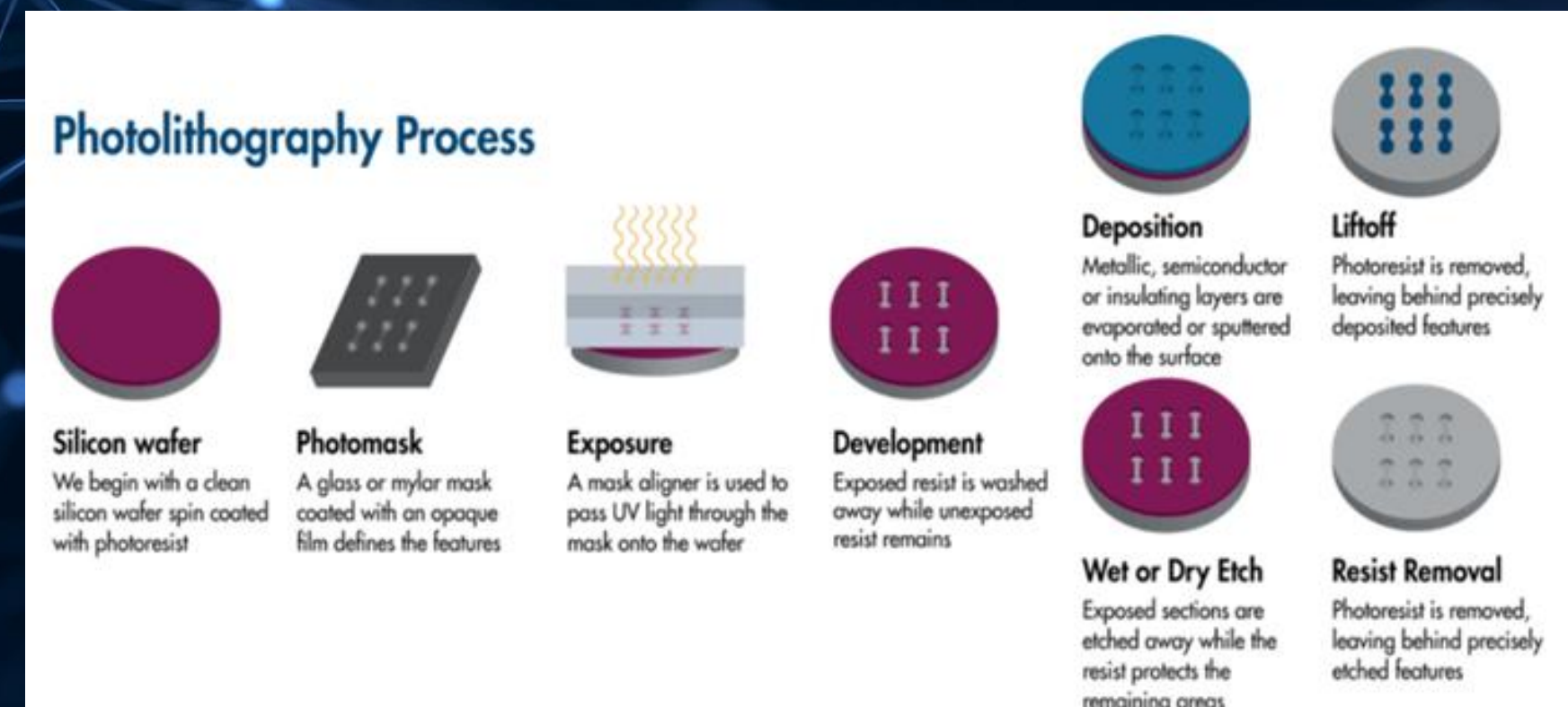
- Description: Uses light to transfer patterns from a photomask to a light-sensitive photoresist on the wafer. The photoresist is exposed, developed, and used as a mask for etching or deposition.

Advantages: High throughput, cost-effective for large-scale production.

Limitations: Limited by diffraction to ~193 nm (using deep ultraviolet, DUV). Extreme ultraviolet (EUV) lithography extends resolution to ~2 nm but is costly.

Process:

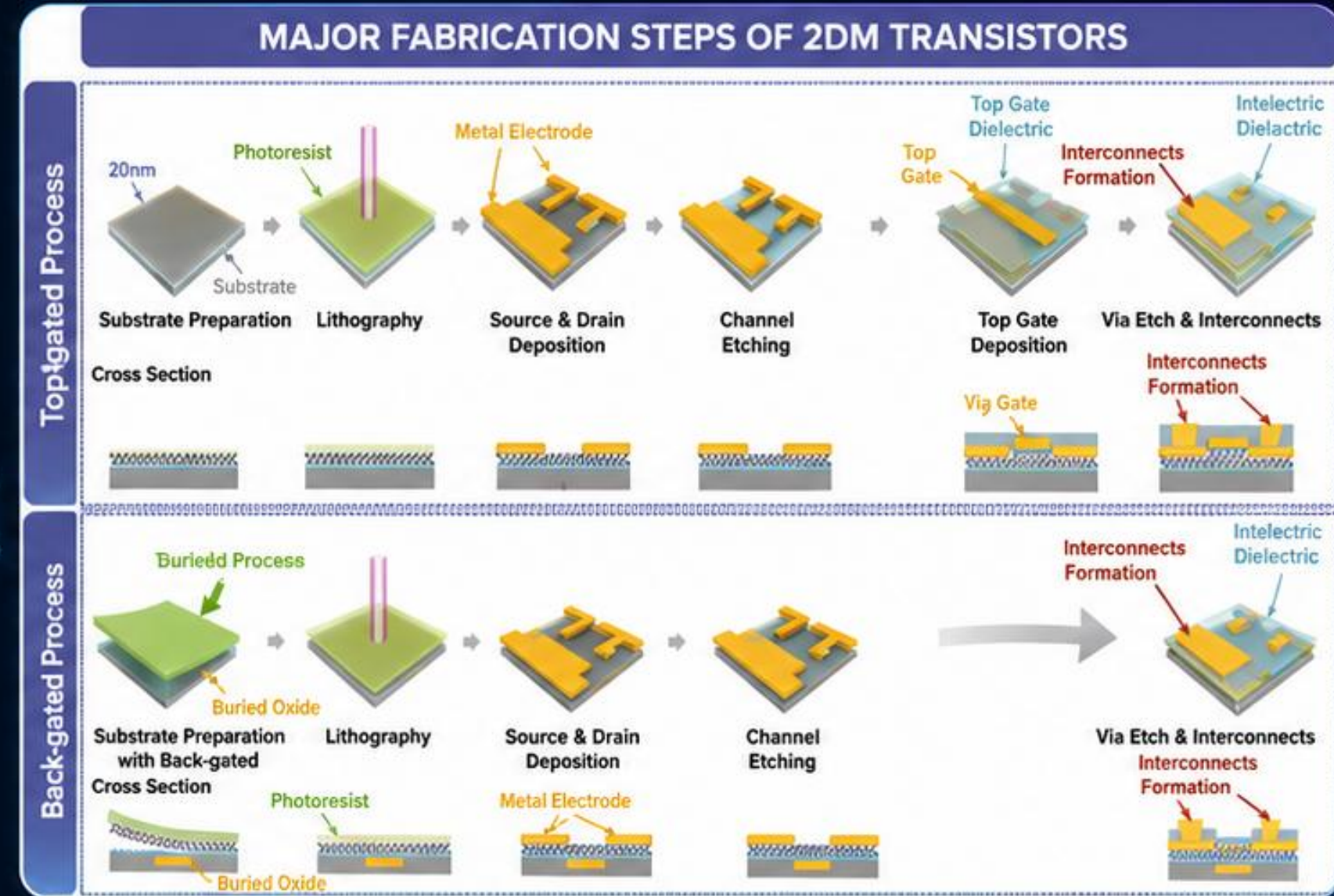
1. Coat wafer with photoresist.
2. Expose photoresist to UV light through a mask.
3. Develop photoresist to reveal the pattern.
4. Etch or deposit material, then remove photoresist



3. MAIN STEPS IN MANUFACTURING A NANOSCALE TRANSISTOR

The fabrication of a nanoscale transistor (e.g ; FinFET) involves multiple steps:

- 1 SUBSTRATE PREPARATION:**
 - Start with a silicon wafer, cleaned and polished.
- 2 FIN FORMATION:**
 - Use photolithography and etching to create 3D silicon fins.
- 3 GATE DIELECTRIC DEPOSITION:**
 - Deposit a high-k dielectric (e.g; HfO_2) using ALD for insulation.
- 4 GATE ELECTRODE FORMATION:**
 - Deposit metal gate (e.g; TiN) using CVD or sputtering.
- 5 SOURCE/DRAIN DOPING:**
 - Implant dopants to form source and drain regions.
- 6 SPACER FORMATION:**
 - Deposit insulating spacers (e.g; SiN) to isolate gate from source/drain.
- 7 INTERCONNECTS:**
 - Deposit metal layers (e.g; copper) for electrical connections.
- 8 TESTING AND PACKAGING:**
 - Test functionality and package the chip.

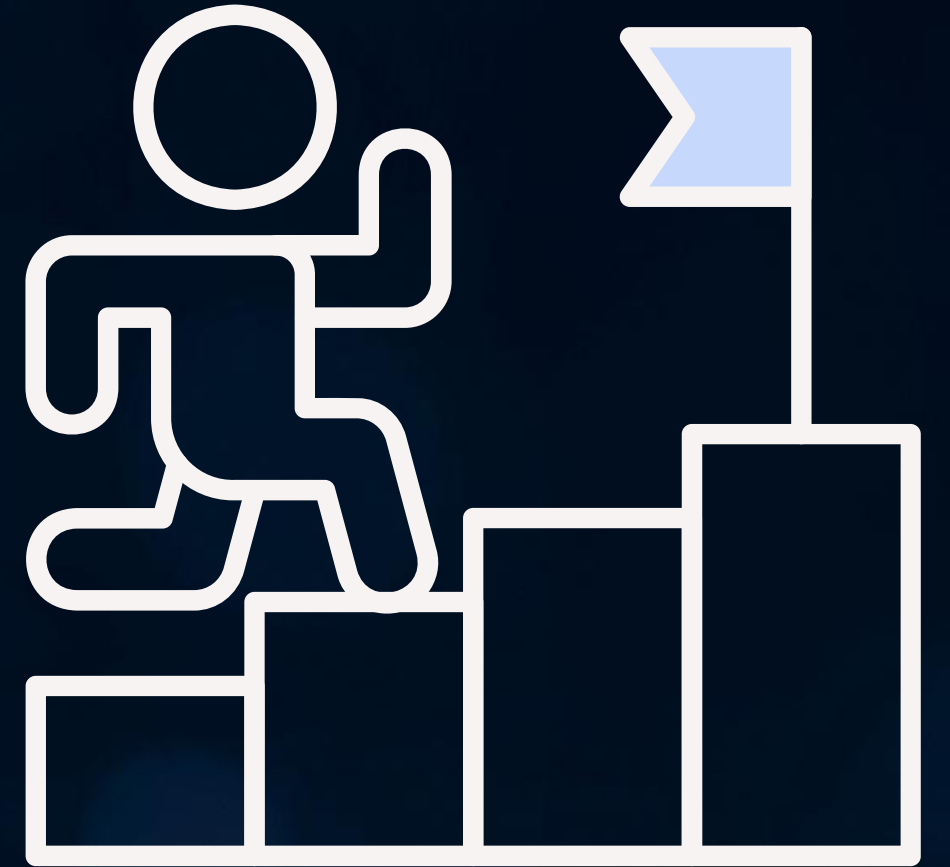


FROM ATOMS TO APPLICATIONS:
Engineering the Future, One Nanometer at a Time.



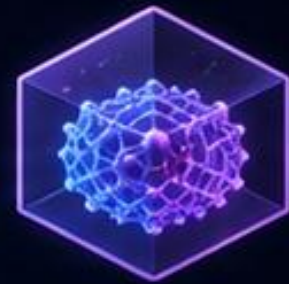
4. CHALLENGES IN NANOSCALE FABRICATION

- **Precision:**
 - Achieving sub-10 nm features requires extreme accuracy. EUV lithography and e-beam lithography are costly and complex.
- **Defects:**
 - Atomic-scale defects (e.g; vacancies, dislocations) can degrade performance. Contamination control is critical.
- **Cost:**
 - Advanced tools (e.g; EUV lithography machines) cost hundreds of millions of dollars. R&D for new processes is expensive.
- **Scalability:**
 - Balancing high precision with high throughput remains a challenge.





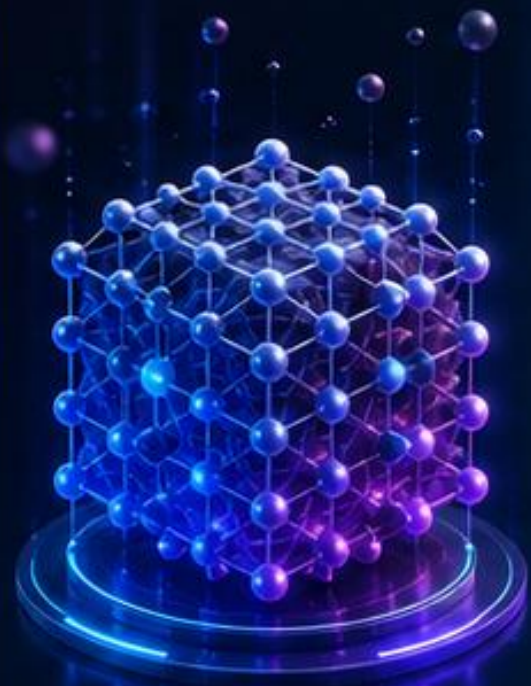
APPLICATION OF SEMICONDUCTOR IN NANODEVICES



OVERVIEW

Semiconductor nanomaterials have interesting physical and chemical properties and useful functionalities, such as narrow and intensive emission, continuous absorption bands, high chemical and photobleaching stability, processability, and surface functionality, which are among the most attractive properties of these materials.

Semiconductor nanomaterials have been the focus of research for about 20 years. The movement of electrons and holes in semiconductor nanomaterials is primarily governed by the well-known quantum effects.



KEY MECHANISMS AND DISCOVERIES:



– Size-tunable electronic bands



– Photobleaching stability through chemical passivations



– Exciton dynamics control



– Novel doping techniques for nanomaterials

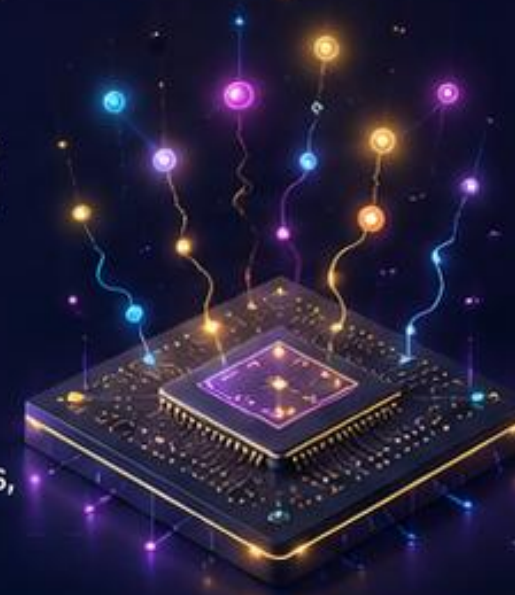


– Recent advances in 2D semiconductor integration

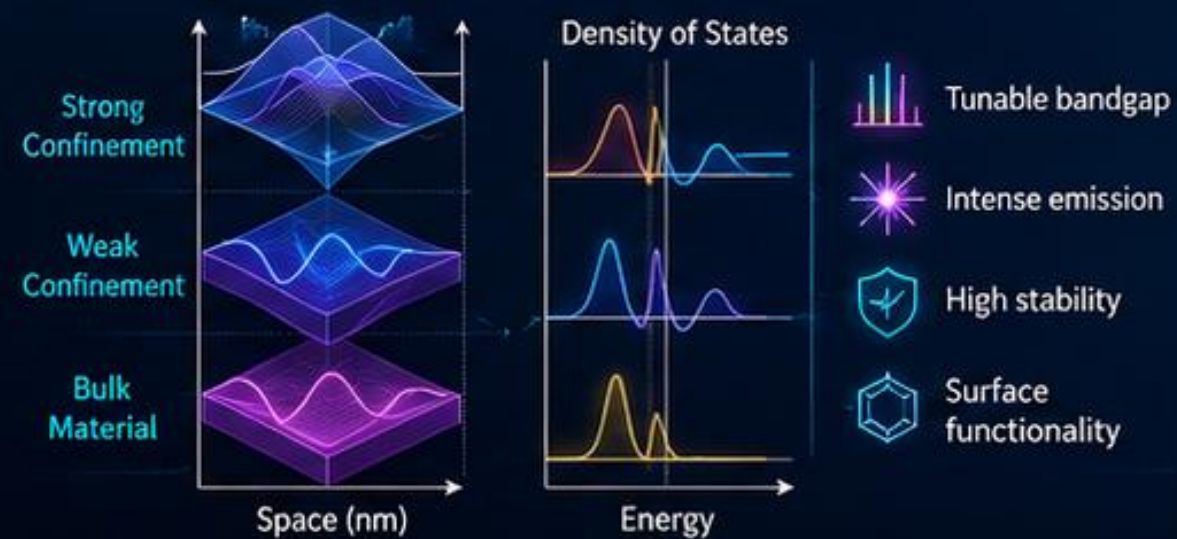


IMPACT & APPLICATIONS

Transport properties related to phonons and photons are largely affected by the size and geometry of the materials. These novel properties have attracted significant attention in research and applications in nanoelectronics, nanophotonics, energy conversion, nonlinear optics, miniature sized sensors and imaging devices, solar cells, catalysis, detectors, photography, and biological sciences and biomedicine.



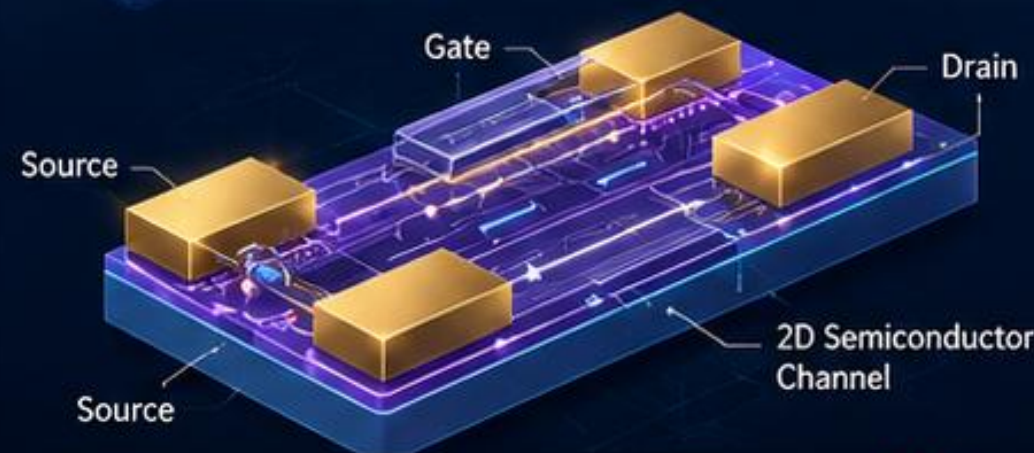
QUANTUM CONFINEMENT EFFECTS



Electron and hole behavior in nanomaterials is governed by quantum confinement and discrete energy levels.



ADVANCED NANODEVICE INTEGRATION



High Performance



Low Power Consumption



Scalable Architecture

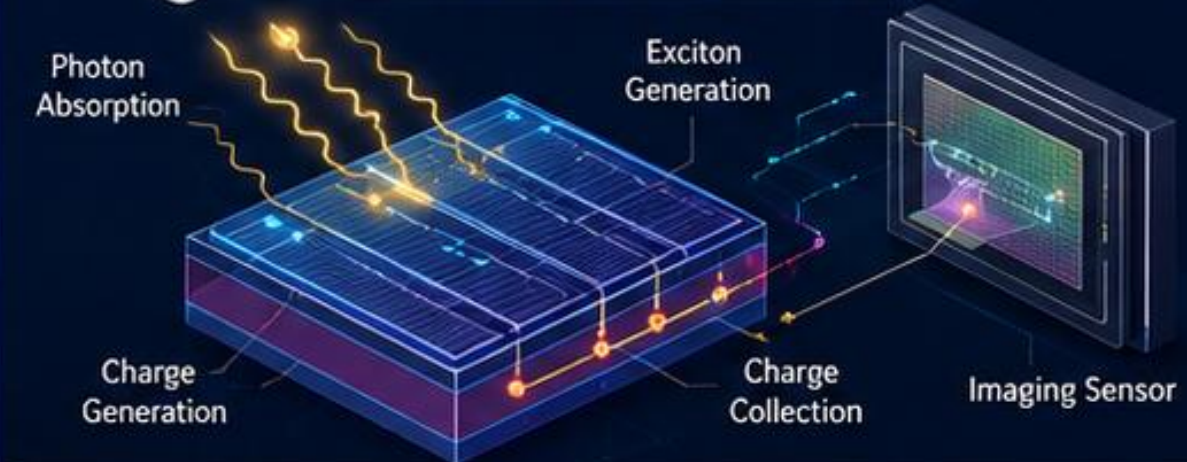


Next-Gen Electronics

Innovative material engineering and 2D integration enable faster, smaller, and more efficient nanoelectronics.



APPLICATIONS ACROSS DISCIPLINES



Nanoelectronics



Nanophotonics



Energy Conversion



Sensors & Imaging



Catalysis



Bio & Medicine

From energy to healthcare, semiconductor nanodevices are shaping a smarter, sustainable future.



Smaller Scale.



Stronger Performance.



Smarter Future.

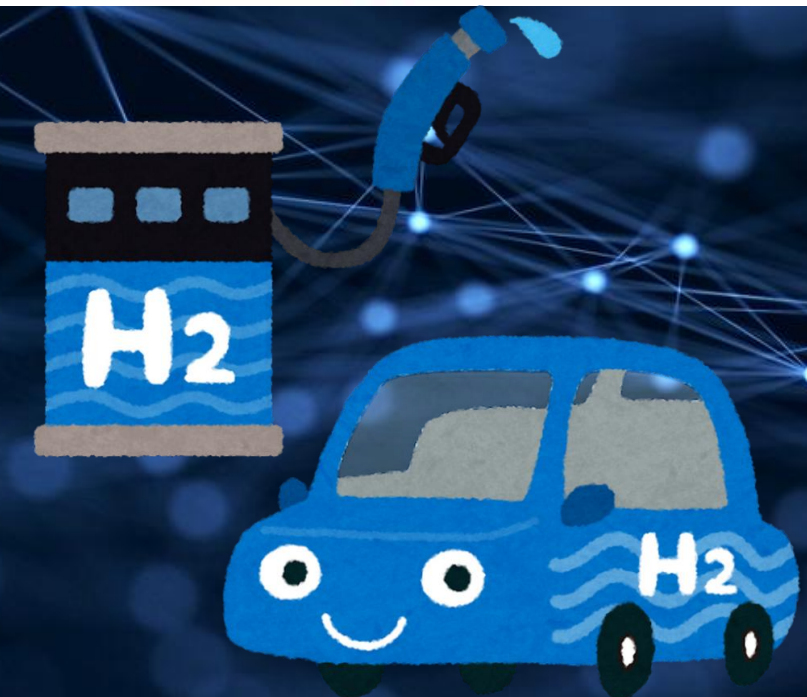
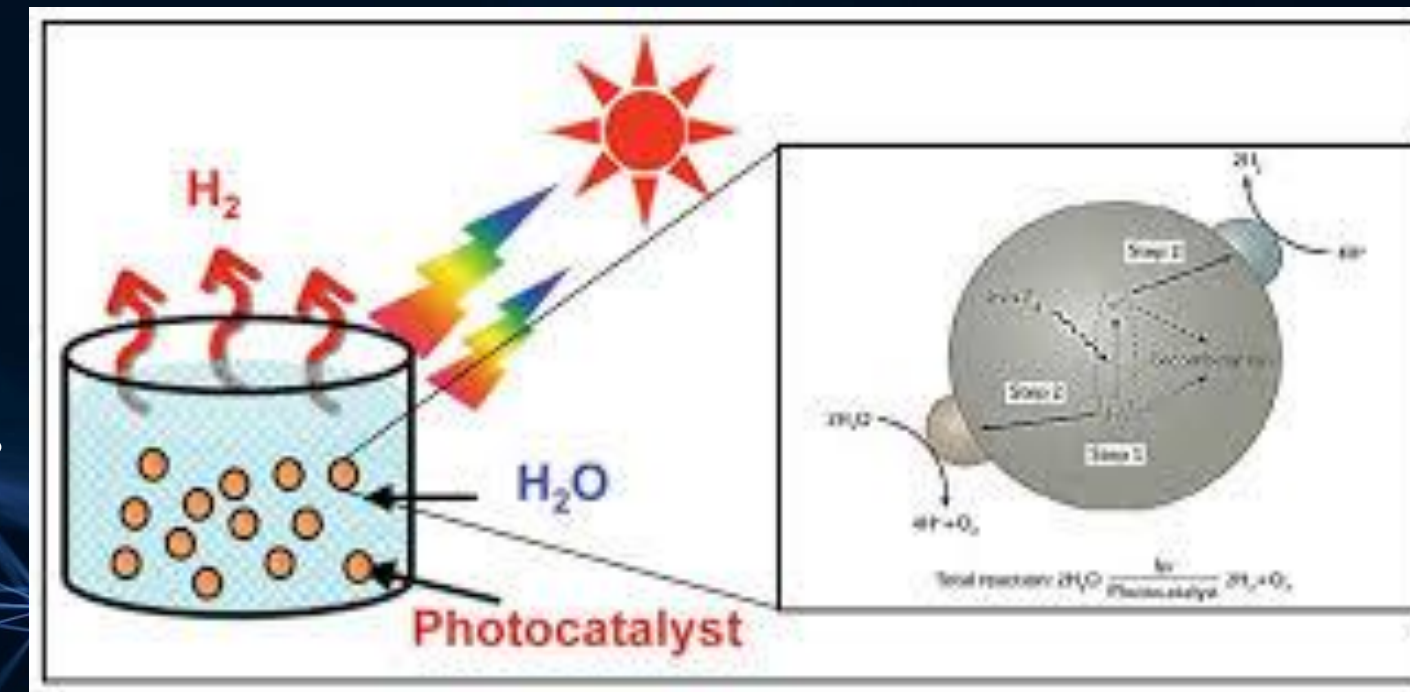
SEMICONDUCTOR NANOMATERIALS FOR HYDROGEN PRODUCTION:

The target market is a group of potential customers for the new product or service in the expansion region. This includes demographics, psychographics, and purchasing behavior. The market size, growth rate, and trends are analyzed. Market research is conducted to understand the market's needs, preferences, and competitive landscape.

Hydrogen is a promising alternative fuel, since it is completely pollution-free and can readily be produced from renewable energy resources. thus eliminating the net production of greenhouse gases.

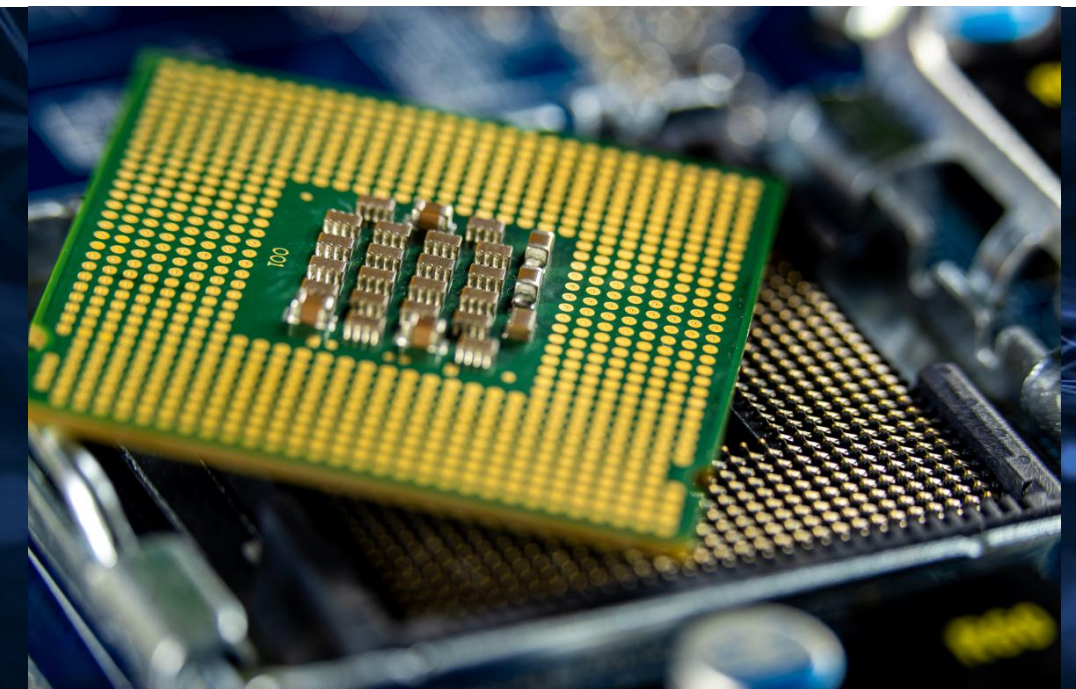
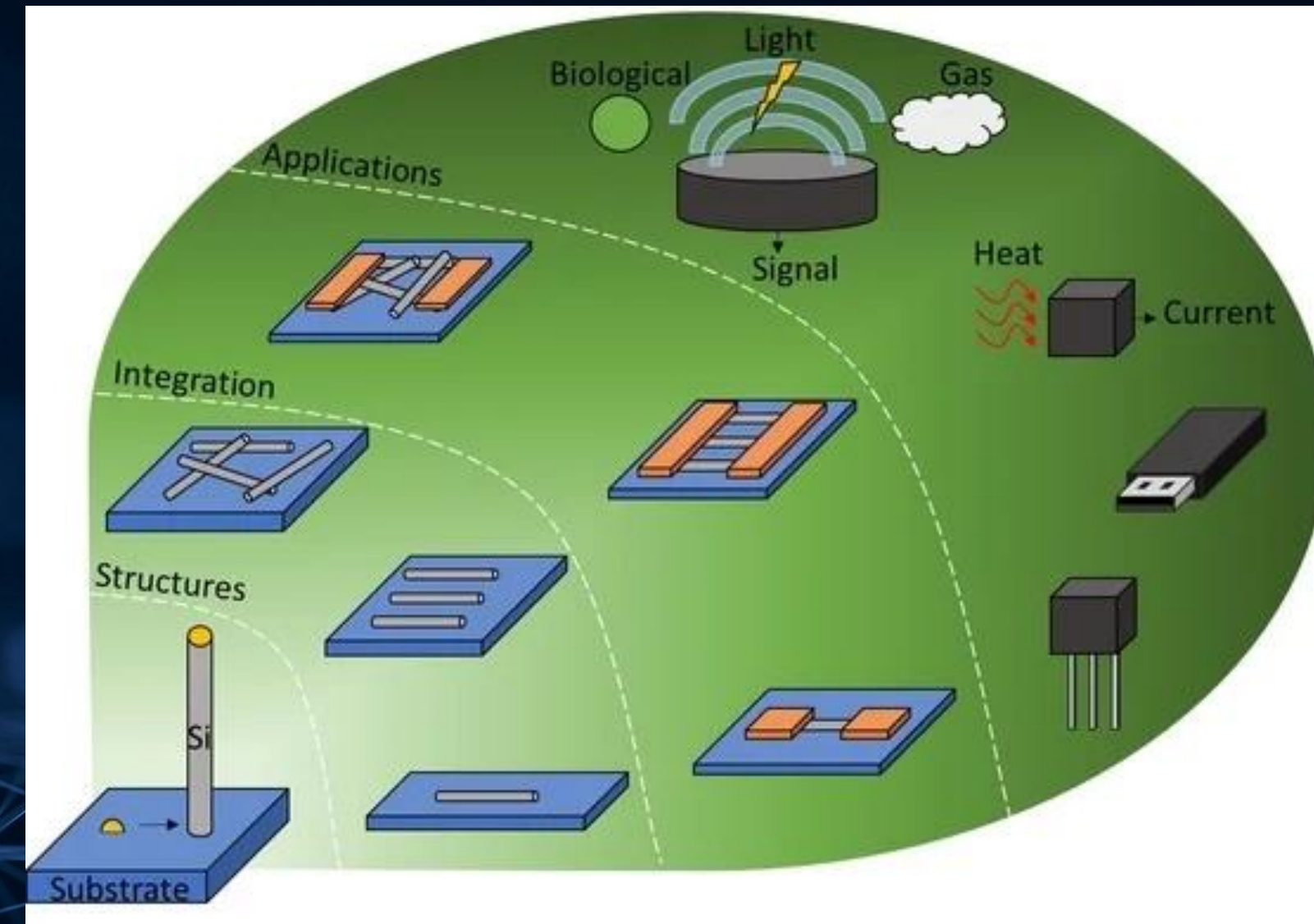
Recent studies have indicated that hydrogen fuel costs are reasonable and hydrogen is therefore an ideal candidate to replace fossil fuels as an energy carrier. The production of hydrogen from liquid hydrocarbon fuels and adsorption on various solids including carbon has attracted a lot of attention by the media and the automotive industry. Photocatalysis is expected to make a great contribution to both environmental treatment (emission cleaning and water purification) and renewable energy. Hydrogen (H₂) is widely considered to be the future clean energy carrier in many applications,

a parallel Swedish study reported that pollution can increase the risk of having cancer by 40-70 %



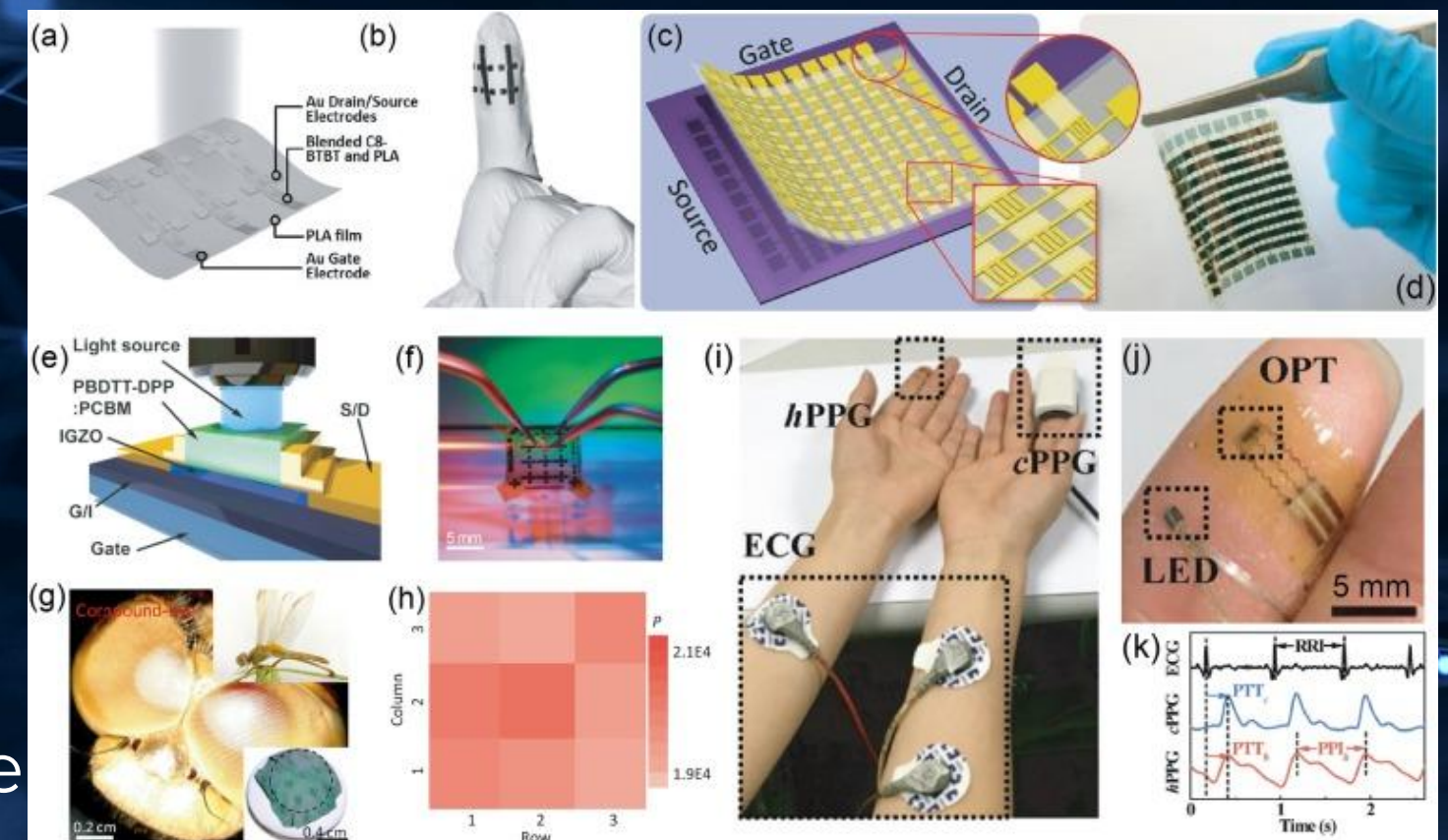
SILICON SEMICONDUCTOR NANOMATERIALS AND DEVICES

This field has become one of the most active research areas of semiconductor nanotechnology. Knowledge of modern information technology is based on silicon based microelectronics. Therefore, the development of silicon based materials and devices will influence the future development of current microelectronics and information technology to a certain extent, which has an important strategic significance. Besides the widely used crystalline silicon wafers, silicon nanostructures have been considered as the basic components for future nano/microelectronic devices. With the size of silicon based complementary metal oxide semiconductors (CMOS) circuit reducing gradually down to 10 nm or even smaller, silicon based nanotechnology can provide a solution with lower cost and higher efficiency. The size effect of silicon nanostructures leads to a lot of novel



ORGANIC OPTOELECTRONIC MATERIALS AND DEVICES

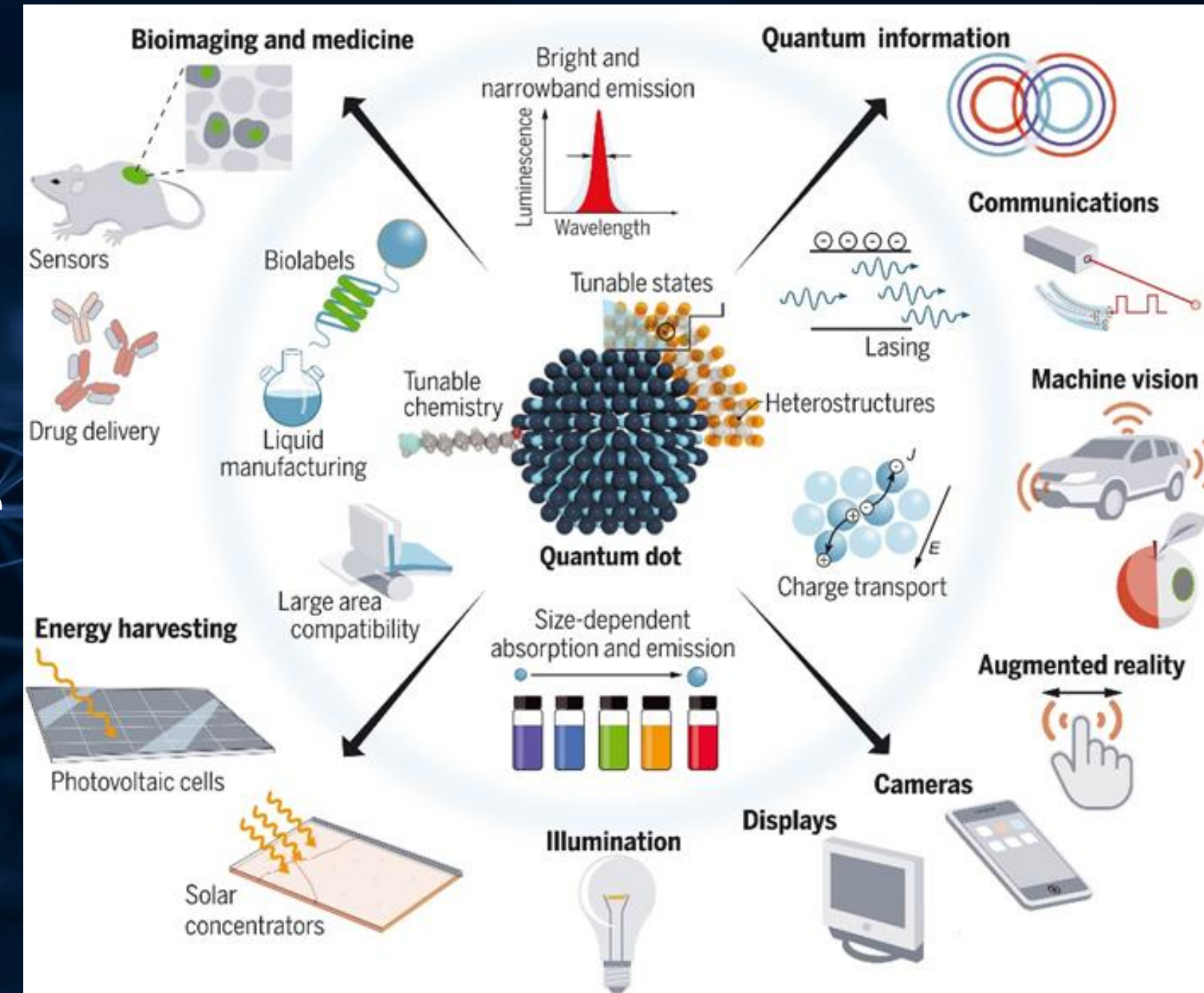
organic optoelectronic materials with strong photoluminescence, nonlinear optical materials, and some important low molecular weight pharmaceutical materials, construct the nanodevices and explore their practical applications. Organic optoelectronics has developed into a new interdisciplinary research field, involving organic chemistry, physics, electronic engineering and materials science. Organic optoelectronic devices, such as organic electroluminescent devices (OLED), organic photovoltaic (OPV) and organic thin film transistors (OTFT), have attracted significant attention in academics and industries, due to their great application potential in flat-panel and flexible display, solid-state lighting, information transport and storage, new generation energy, photocatalyst and so on. Owing to the advantages of solid-state, self-emission, full color capability and flexibility, OLED has been recognized as one of the most promising flat-panel display technologies and has just become commercial.



SEMICONDUCTOR INNOVATIONS AND CHALLENGES

quantum dots

Semiconductor materials feature optical and electronic properties that can be engineered through their composition and crystal structure. The use of semiconductors such as silicon gallium arsenide sparked technologies from computers and mobile phones to lasers and satellites. Semiconductor quantum dots (QDs) offer an additional lever: Because their size is reduced to the nanometer scale in all three dimensions, the restricted electron motion leads to a discrete atom-like electronic structure and size-dependent energy levels. This enables the design of nanomaterials with widely tunable light absorption, bright emission of pure colors, control over electronic transport, and a wide tuning of chemical and physical functions because of their large surface-to-volume ratio.



THE ENVIRONMENTAL CHALLENGE OF SEMICONDUCTORS

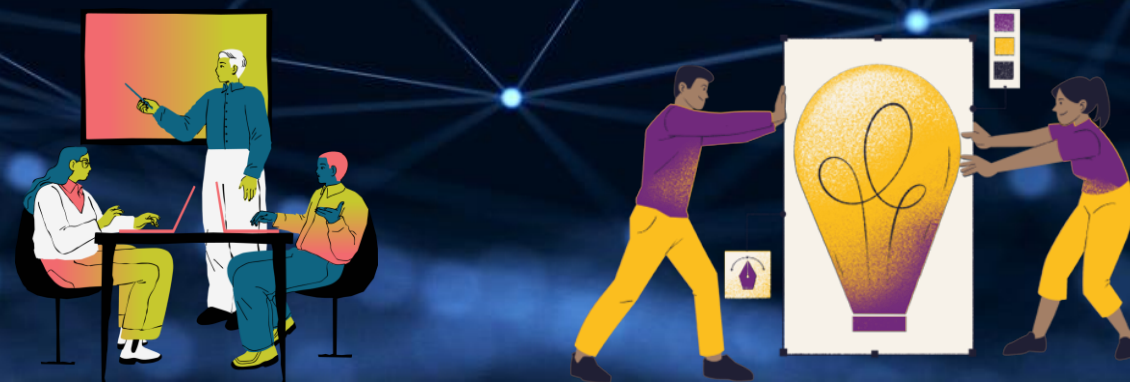
Semiconductor manufacturing has long been associated with resource-intensive processes, high energy consumption, and the generation of hazardous waste. As global demand for electronic devices continues to grow, so does the pressure to minimize the industry's carbon footprint and environmental impact.

Energy Efficiency in Manufacturing

One of the most significant contributors to the environmental impact of semiconductor manufacturing is its high energy consumption. To mitigate this, semiconductor companies are investing in cutting-edge technologies and processes that improve energy efficiency throughout the manufacturing cycle.

- Advanced Materials

The use of advanced materials can significantly enhance energy efficiency. For example, silicon carbide (SiC) and gallium nitride (GaN) are wide-bandgap materials that enable more energy-efficient power devices, reducing energy losses and improving overall performance.



Waste Reduction and Recycling

1- Chemical Recycling

Chemical recycling processes are being developed to recover and reuse valuable materials from semiconductor manufacturing waste. These processes can reduce the need for new raw materials, lower waste disposal costs, and decrease the environmental impact of the industry.

2- Ultrasonic Cleaning

Ultrasonic cleaning technology uses high-frequency sound waves to clean semiconductor components more efficiently. This reduces the amount of water and chemicals required for cleaning, making the process more sustainable.

SOME SCIENTIST WHO GOT A PRIZE IN SEMICONDUCTORS RESEARCH UCLA



Kang Wang

Distinguished Professor and
Raytheon Chair in Electrical
Engineering
Primary Area: Physical and
Wave Electronics



**Benjamin
Williams**

Professor
Primary Area: Physical and Wave
Electronics



Danijela Cabric

Professor
**Primary Area: Signals &
Systems**

UCLA ENGINEERS RECEIVE THREE NSF AWARDS FOR SEMICONDUCTOR RESEARCH

three research teams led by the UCLA Samueli School of Engineering have been awarded research grants totaling \$3.38 million by the National Science Foundation (NSF) through its [Future of Semiconductors design and manufacturing program](#). Electrical and computer engineering distinguished professor [Kang Wang](#), as well as professors and vice chairs [Benjamin Williams](#) and [Danijela Cabric](#), lead the three projects, respectively.

Williams' group has been awarded a three-year, \$1.96 million grant to develop hybrid electronic-photon semiconductor devices that operate at frequencies above 1 terahertz (THz). Current semiconductors fabricated at silicon-industry foundries can generate and detect radiation at ever-increasing frequencies

CONCLUSION

Semiconductors are essential for nanodevices, enabling high-performance electronics, sensors, and optoelectronics. Their tunable properties (e.g., bandgap, mobility) make them ideal for transistors, LEDs, and quantum devices. Challenges like quantum effects and heat dissipation remain, but advancements in nanomaterials (quantum dots, nanowires, 2D materials) promise faster, smaller, and more efficient technologies. Future progress will drive innovations in computing, healthcare, and energy.

CREDITS



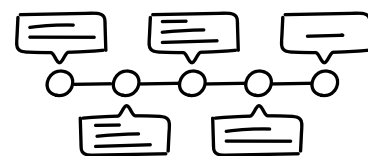
eng. Hassan emad
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the leader of the
group and made the
power point
and made
(application of
semiconductor)



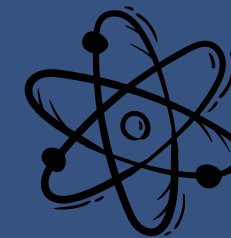
eng. Amira Ibrahim

made
(semiconductor
materials for
nanodevices) part



**eng. mohamed
khald**

made
(fabrication methods
for semiconductor in
nanodevices)
part



eng. Alzahraa Refaat

made
(**fundamental physics
of semiconductor**)
part



**eng. Mohamed
mostafa**

made
(innovation and ethical
responsibility in the future of
Nano semiconductors)



THANKS FOR LISTENING

do you have any question?